Week 20: From Six Strong Examples to Your Research Proposal

Find Patterns, Design a Project, Write the Story, and Present It Clearly

Instructor / Mentor

Kiki Research Training Program

Week 20 of the Research Training Program

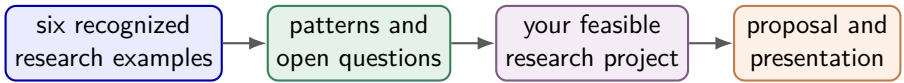
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Today's Mission

Week 20 Is a Synthesis Workshop

Today we do not add a new technical algorithm. We connect what you have already learned.



Main deliverable during class

Finish and improve the Week 18 six-material comparison table, then use it to outline one research project.

Learning Objectives

By the end of class, students can:

1. complete every row of the six-material research comparison table;
2. distinguish a real cross-source trend from a repeated buzzword;
3. identify common grid problems, research questions, methods, simulators, and data strategies;
4. turn an interest into a bounded research question and an evidence plan;
5. organize a short research proposal; and
6. present a project as one connected argument rather than a list of facts.

120-Minute Flow

1. **0–10 min:** proposal puzzle—what makes a project complete?
2. **10–30 min:** finish the six-source table row by row.
3. **30–47 min:** find defensible trends across the six sources.
4. **47–52 min:** break.
5. **52–70 min:** map problems, questions, methods, tools, and data.
6. **70–88 min:** use TRIZ to generate and improve project ideas.
7. **88–103 min:** build a one-page proposal framework.
8. **103–115 min:** practice a three-minute presentation and peer review.
9. **115–120 min:** revise one weak link and launch the short homework.

Opening Puzzle: Is This a Research Project?

Consider the statement:

> *"I want to use AI to make the smart grid better."*

In pairs, identify what is missing. Try to produce at least five questions.

A topic is not yet a project

A research project needs a specific problem, stakeholder or context, research question, evidence, method, comparison, success measure, boundary, and feasible output.

Finish the Six-Material Review

The Six Sources Have Different Jobs

Your packet contains two groups:

Three NSF award abstracts

- describe planned work;
- emphasize objectives and significance;
- name expected methods and impacts;
- do not prove that the expected result occurred.

Three published papers

- report completed analysis or experiments;
- provide methods, results, and limitations;
- support claims with evidence;
- may inspire a new question rather than a direct copy.

Tense rule

For an award abstract, write “proposes,” “plans,” or “aims.” For a paper, write “tests,” “reports,” or “finds.”

How to Fill Each Row: Title and Source

Record enough information that another student could locate the material:

- complete title;
- lead investigator or first author;
- NSF award number or journal name;
- year;
- source type: award abstract or published paper.

Strong entry

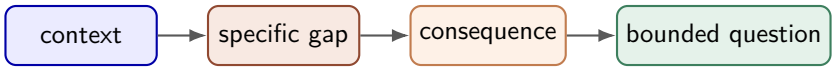
"[Title], [investigator/author], NSF Award [number], [year]—award abstract."

Weak entry

"AI grid paper." It cannot be checked and loses the source's identity.

How to Fill Each Row: Problem and Research Question

Separate the broad domain from the exact gap.



Sentence frame

In [context], existing [system/method/data] cannot reliably [task], which matters because [consequence]. The work asks whether/how [specific change] improves [measurable outcome].

How to Fill Each Row: Methodology

"Uses AI" is not a methodology. Extract the complete logic:

1. **Input:** data, measurements, simulations, documents, or participants;
2. **Operation:** model, experiment, optimization, control rule, or analysis;
3. **Comparison:** baseline, control group, traditional method, or ablation;
4. **Output:** prediction, decision, design, explanation, or measurement;
5. **Evaluation:** metric, stress test, uncertainty, and decision rule.

Method sentence frame

The researchers apply [operation] to [input], compare it with [baseline], and evaluate [output] using [metric/test].

How to Fill Each Row: Data and Resources

Look for resources in four categories:

Category	Questions to answer
Data	What variables, locations, time range, labels, and sampling rate?
Tools	Which simulator, programming language, model library, sensor, or laboratory?
People	Which expertise, community partner, participant, or decision maker?
Access	Is it public, licensed, private, costly, sensitive, or difficult to reproduce?

Feasibility test

If you cannot name how a student would obtain or create the input, the project plan is incomplete.

How to Fill Each Row: Results and Expected Outcomes

NSF award abstract

Record the expected capability, product, knowledge, educational activity, or societal contribution. Label it **planned**, not proven.

Published paper

Record the main observed result, the comparison supporting it, and an important limitation or uncertainty.

Do not collapse three ideas

"The model produced an output," "the output was accurate," and "the output would improve the real grid" are three different claims requiring different evidence.

How to Fill Each Row: Impact, Community, and Inspiration

Ask three levels of questions:

1. **Impact:** What technical or social condition could improve?
2. **Community:** Who benefits, contributes data or labor, makes decisions, or could face risk?
3. **Inspiration:** What research move can I adapt without copying the entire project?

Useful inspiration is specific

"I can compare a physics-based baseline with an ML model under missing-sensor conditions" is stronger than "I also want to use AI."

Use Evidence Labels While Completing the Table

Put one label beside each important entry:

TEXT The material states it directly. Record a page, paragraph, figure, or sentence.

INFERENCE It is a reasonable interpretation, but the source does not say it directly.

QUESTION The short material does not provide enough information.

Good researchers preserve uncertainty

A blank or a QUESTION label is more honest than inventing a method, dataset, result, or community partnership.

Find Trends Across Six Sources

What Counts as a Trend?

A trend is a repeated, meaningful pattern supported by multiple sources.

Not enough	Stronger trend statement
"Four sources say AI."	"Four sources combine data-driven models with physical constraints, simulation, or domain measurements."
"Reliability appears often."	"Five sources evaluate failures, disturbances, uncertainty, or unequal service rather than only average performance."
"They use data."	"Several sources respond to scarce operational data through sensing, open benchmarks, or simulation."

Minimum evidence rule

Name the sources, state what they share, and also note an important difference.

A Simple Trend-Finding Procedure

1. Highlight repeated nouns: reliability, renewables, uncertainty, equity, privacy, resilience.
2. Highlight repeated actions: predict, optimize, detect, control, measure, simulate.
3. Group related terms into one idea.
4. Count which sources support the idea.
5. Compare how award abstracts plan to address it and how papers evaluate it.
6. Write one exception, difference, or missing perspective.

Trend sentence frame

Across Sources [labels], researchers increasingly [shared action] because [shared pressure]. They differ in [method/context/evidence], and the packet leaves [open issue] unresolved.

Candidate Trends to Test Against the Packet

These are **hypotheses**, not automatic answers. Keep only those supported by the six materials.

- movement from average performance toward resilience during rare or changing conditions;
- integration of renewable energy, storage, electric vehicles, and flexible demand;
- hybrid use of physics, optimization, simulation, and machine learning;
- growing importance of open, synthetic, or privacy-preserving data;
- evaluation beyond accuracy: safety, constraint violations, robustness, fairness, and interpretability;
- closer attention to who receives reliable electricity and who bears system costs or risks.

The Recurring Big Problem

Many smart-grid projects can be connected to one large challenge:

How can a changing electricity system remain reliable, affordable, sustainable, secure, and fair when supply, demand, devices, weather, and human behavior are uncertain?

A student project must choose only one small part:

- one location or test network;
- one decision or prediction task;
- one time scale;
- one failure or uncertainty;
- one stakeholder;
- one measurable outcome.

Research-Question Families in This Domain

Family	Typical question	Possible evidence
Forecasting	What will load, solar output, price, or risk be later?	chronological test; naive baseline; MAE/RMSE; peak errors
Detection	Is this measurement or event abnormal?	labeled or injected events; precision/recall; false alarms
Optimization	What decision best meets goals and constraints?	cost, emissions, unmet demand, voltage/line violations
Control	What action keeps the system safe as conditions change?	simulated disturbances; stability; recovery time
Planning	What assets or policies are needed over months or years?	scenarios; sensitivity; investment and reliability outcomes
Measurement/equity	What important experience is hidden by current metrics?	sensor/community data; subgroup comparison; uncertainty
Trustworthy AI	When does an AI method fail or violate physics?	stress tests; out-of-distribution cases; calibration; explanations

Methods, Simulators, and Synthetic Data

Common Method Families

Choose a method because it matches the question—not because it sounds advanced.

- **Physical modeling:** power flow, dynamic models, graph/network equations.
- **Statistical and ML prediction:** regression, tree models, neural networks, time-series models.
- **Optimization:** dispatch, scheduling, placement, sizing, and constrained decision making.
- **Detection and classification:** faults, cyberattacks, equipment states, and anomalies.
- **Control and reinforcement learning:** sequential actions with feedback.
- **Human-centered and causal methods:** experiments, surveys, interviews, field measurements, and policy evaluation.
- **Hybrid methods:** physics-informed learning, co-simulation, and model-plus-data approaches.

Simulator and Tool Map

Tool	Best suited to	Student-scale use
pandapower	Python-based power flow, optimal power flow, state estimation, short circuit	small IEEE-style network; vary load or DER; export bus/line results
MATPOWER	MATLAB/Octave steady-state power flow and optimization	compare AC/DC power flow or generation-cost decisions
OpenDSS	detailed electric distribution-system simulation	voltage profiles, solar/EV penetration, faults, time-series feeder studies
GridLAB-D	distribution systems, end-use loads, DERs, markets, time evolution	simulate homes/devices and grid interactions across time
PyPSA	multi-period power and energy-system optimization	generation, storage, transmission, and renewable scenarios
Python data tools	NumPy, pandas, and scikit-learn for data generation, cleaning, baselines, and ML	turn simulator output into a reproducible analysis pipeline

Selection rule

Start with the simplest tool that represents the variables and constraints required by the research question.

How to Create a Synthetic Dataset Responsibly

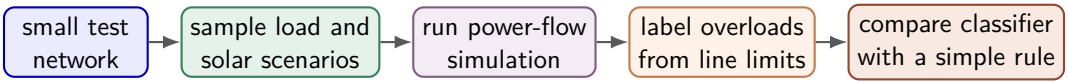
1. Choose a documented test network or build a small transparent network.
2. Define normal operating ranges and units.
3. Vary one or more factors: load, weather, solar, EV charging, outages, sensor noise, or attacks.
4. Run the simulator and save both inputs and outputs.
5. Create labels from known injected events, not guesses made afterward.
6. Split scenarios so the test set contains unseen times or conditions.
7. Compare distributions and operating constraints with a real or published reference when possible.
8. Document every assumption, random seed, version, and limitation.

Boundary

Synthetic data can test logic and rare scenarios. It does not automatically prove performance on a real utility system.

A Small Synthetic-Data Project Example

Question: Can a classifier detect overloaded lines when demand and rooftop solar change?



Metrics: recall for overloads, false-alarm rate, performance on unseen demand ranges, and number of physical constraint violations.

Use TRIZ to Form a Project

TRIZ: Structured Inventive Problem Solving

TRIZ is commonly translated as the **Theory of Inventive Problem Solving**. It treats innovation as a process that can be supported by reusable patterns.
For this course, use four simple ideas:

1. describe the system and the desired outcome;
2. identify a contradiction—improving one property seems to worsen another;
3. search for available resources already inside or around the system; and
4. use inventive principles as prompts for alternatives, then test them with evidence.

TRIZ does not replace research

It helps generate possible directions. Experiments, data, simulation, and analysis decide whether an idea works.

Find the Contradiction

Many research ideas begin with a tension:

We want more...	but	we also want...
renewable-energy use		reliable operation during variability
detailed data		privacy and low sensor cost
accurate AI models		interpretability and fast computation
local control		coordination across the whole grid
low electricity cost		resilience and fair service

Project seed

How can we improve [desired property] without unacceptable loss of [competing property], under [specific condition]?

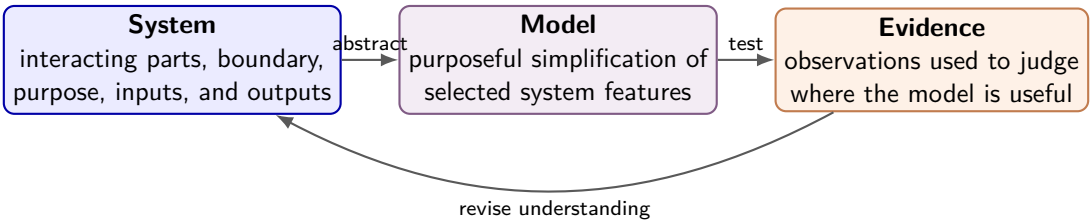
TRIZ Prompts for a Smart-Grid Idea

Use these simplified prompts to produce alternatives:

- **Segment:** divide the grid, data, users, or time periods into meaningful parts.
- **Act locally:** use different rules for different nodes or conditions.
- **Prepare beforehand:** forecast, reserve, pre-charge, or pre-position before a peak or fault.
- **Use feedback:** measure the outcome and adjust the next action.
- **Add a mediator:** introduce storage, an aggregator, a digital twin, or an explanation layer.
- **Turn harm into information:** use failures, residuals, or constraint violations to reveal a research question.

Generate at least three alternatives before choosing one.

System, Model, and Reality



Every model leaves something out

The important question is not "Is this model perfectly real?" but "Is it adequate for this question, scale, and decision?"

Write and Present the Proposal

The Project Builder Table

Row	What you must fill in
Working title	object of study + action/question + context
Motivation	observation or personal reason; why you care
System boundary	included parts, excluded parts, time scale, and location
Problem/gap	what is unknown, unreliable, unavailable, unsafe, or unfair
Stakeholder	who uses, experiences, decides, benefits, or bears risk
Research question	one bounded question with a measurable outcome
Hypothesis	expected relationship and reason; optional for exploratory work
Data/resources	accessible data, simulator, software, people, computing, permissions
Method	input, operation, comparison, output, and evaluation
Baseline	simplest meaningful alternative
Success measure	metric, threshold, stress test, or decision rule
Minimum output	smallest honest project that can be completed on time
Limitations/ethics	what the study cannot claim; privacy, fairness, safety, or access
Broader impact	beneficiary, pathway to benefit, and how benefit would be checked

A Strong Research Question Has Five Parts

Template

Under [context/condition], how does [method or change] affect [measurable outcome] compared with [baseline] for [system/stakeholder]?

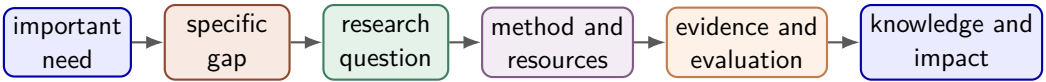
Example

Under unseen high-solar scenarios in a small distribution-network simulation, how does a physics-informed overload classifier affect overload recall and false alarms compared with logistic regression?

Scope check

If the question requires private utility data, unavailable hardware, many years, or proof about every grid, shrink it before selecting the model.

The Proposal Is a Chain of Reasoning



Review every arrow. A proposal is weak when two neighboring boxes do not logically connect.

A Student-Scale Proposal Structure

1. **Title and overview:** the project in 3–5 sentences.
2. **Background and gap:** what is known, what is missing, and why it matters.
3. **Research question and objective:** exactly what the project will determine or create.
4. **Method:** data, system/model, procedure, baseline, and metrics.
5. **Feasibility:** tools, access, schedule, and minimum complete output.
6. **Expected outcomes:** possible results and what each would mean.
7. **Limitations, risks, and ethics:** what could fail and what will not be claimed.
8. **Intellectual merit and broader impact:** new understanding and pathway to benefit.
9. **References:** sources for facts, methods, data, and prior work.

Do Not Promise the Result

A proposal requests permission and resources to learn something. The conclusion is not known yet.

Weak promise	Research-ready statement
"Our AI will make the grid reliable."	"We will test whether the method reduces missed overloads under defined simulated scenarios."
"The model will be better."	"We will compare both methods using the same train/test cases and report uncertainty."
"This will help everyone."	"We will identify a specific user and evaluate a plausible pathway to benefit."

The Three-Minute Research Pitch

Use six moves, about 25–30 seconds each:

1. **Hook:** one situation that makes the problem visible.
2. **Gap:** what current knowledge, data, or method cannot do.
3. **Question:** the exact question you will answer.
4. **Plan:** data/model, method, baseline, and evaluation.
5. **Feasibility and honesty:** minimum output, largest risk, and boundary.
6. **Why it matters:** new knowledge and who could use it.

One visual rule

Each slide should do one job. Prefer a diagram, comparison, or result to a paragraph of tiny text.

Peer Review: Test the Weakest Link

After each pitch, reviewers answer:

1. Can I repeat the research question accurately?
2. Does the method produce evidence that answers that question?
3. Is there a meaningful baseline or comparison?
4. Are the data, simulator, and tools actually accessible?
5. Is the evaluation broader than one accuracy number?
6. What is the strongest unsupported assumption?
7. What should be removed to make the project finishable?

Feedback rule

Give one specific strength, one evidence-based concern, and one actionable revision.

Final In-Class Revision

Circle the weakest connection in your proposal chain and revise only that connection:

- problem → research question;
- question → method;
- method → evidence;
- evidence → conclusion;
- conclusion → broader impact; or
- project scope → available time and resources.

Finish this sentence:

My proposal became more credible when I changed _____ *because* _____.

Homework and Resources

Week 20 Homework: Keep It Simple

Answer in your own words. Examples are encouraged.

1. What is TRIZ? Explain its purpose and describe one TRIZ idea that could help someone improve a research project.
2. What is a system, based on your understanding? Give one example and identify its boundary, parts, inputs, interactions, and outputs.
3. What is a model, based on your understanding? Explain how a model differs from the system it represents, what it leaves out, and how we can test whether it is useful.

Suggested length

About 100–150 words for each answer. No coding is required.

Official Tool References

- pandapower: Python power-system modeling, analysis, and optimization
- MATPOWER: steady-state power-system simulation and optimization
- OpenDSS: electric distribution-system simulation
- GridLAB-D: distribution, end-use, DER, and market simulation
- PyPSA: multi-period power and energy-system optimization
- NREL grid modeling tools and test resources
- NSF proposal preparation and merit-review guidance

Use official documentation first

Record the software version and cite the exact test network, dataset, example, or documentation page used in the project.

Final Takeaways

1. Six examples are useful only when you compare them systematically.
2. A trend requires repeated evidence and an acknowledged difference or exception.
3. A topic becomes a project when it has a bounded question, evidence plan, baseline, and feasible output.
4. Simulators can create controlled and rare scenarios, but synthetic evidence has clear limits.
5. TRIZ helps reveal contradictions and generate alternatives; research tests those alternatives.
6. A model is a purposeful simplification of a system, not the system itself.
7. A strong proposal and presentation tell one connected, honest story.

Read across. Think in systems. Test one clear idea.

Your goal is not the most complicated project. It is the clearest complete research argument you can test honestly.